Flooding the filter: a game-theoretic model of AI, submission pressure, and quality collapse

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Abstract

Generative AI is transforming the nature of academic publishing. By September 2024, more than one-fifth of computer-science preprints showed evidence of large-language-model modification, while monthly submissions to arXiv reached a record 24,226, roughly 50% higher than in late 2022. Verification has not kept pace: reviewer acceptance rates have declined steadily over two decades, and between 6.5% and 16.9% of conference peer reviews are now themselves AI-modified. We study this asymmetry—the fact that AI scales paper production faster than it scales screening—using a game-theoretic model in which AI adoption raises submission volume and lowers drafting cost faster than it raises review capacity, while also increasing the rate at which low-quality manuscripts pass screening. We find three results. First, extreme slot scarcity can stabilize honest submission norms even under degraded screening, because the publication lottery becomes too hard to win. Second, gradual AI adoption can trigger a discrete tipping point into a low-quality long-run outcome; once tipped, the system is self-sustaining, so recovery requires a large coordinated intervention, not incremental reform. Third, expanding publication capacity without scaling verification accelerates the tipping. These results point to a concrete policy agenda of verification-centric institutions (structured audits, submission quotas, and decoupled certification) rather than simply more publication slots.

1 Introduction

We ask what happens to a publishing *system* when it becomes substantially cheaper and faster to produce a manuscript than to review one. By September 2024, more than one-fifth of computer-science preprints on arXiv showed measurable evidence of large-language-model modification (Liang et al., 2025), and at least 13.5% of all biomedical abstracts published that year contained AI-assisted text (Kobak et al., 2025). Monthly submissions to arXiv reached a record 24,226 in October 2024, roughly 50% higher than in late 2022 (arXiv, 2024). Verification has not kept pace. Reviewer acceptance rates have declined steadily over two decades (Meyerson et al., 2025), and between 6.5% and 16.9% of conference peer reviews now contain substantial AI-generated text (Liang et al., 2024). More than half of active researchers surveyed in 2025 reported using AI tools to write reviews, often against explicit journal policy (Naddaf, 2025). The question is not whether individual researchers behave ethically, but whether the system as a whole can absorb an asymmetric shock to production and verification.

While the conventional response to AI in science focuses on disclosure requirements, detection tools, and editorial guidelines, these measures address individual conduct and leave untouched a structural problem: even if every author is honest, reviewers can scrutinize only a finite number of manuscripts carefully per cycle. When submission volume rises faster than review capacity, screening degrades—not because of misconduct, but because of congestion. A reviewer who faces twice as many papers in the same time cannot give each submission the same attention. Congested screening lowers the effective quality threshold, and a lower threshold raises the expected payoff from submitting weak or low-quality work. We focus on this feedback between volume and screening quality as the central mechanism.

We distinguish two bottlenecks in the publishing system, and we argue that AI affects them asymmetrically. The first is verification capacity: the number of papers that reviewers can evaluate carefully in a given period, denoted K_{rev}. The second is allocation capacity: the number of papers the journal can publish per period, denoted K. AI raises submission volume V by reducing the marginal cost of drafting

and resubmitting papers, lowering the effort cost c of each submission. It may also increase the false-positive rate λ—the probability that a low-quality manuscript passes peer review—because AI-assisted writing produces polished text that is harder to distinguish from genuine contributions on surface features alone. Review capacity, by contrast, does not scale symmetrically. Deep expertise, data and code verification, and replication checking remain human-intensive tasks, and journal policies in many fields explicitly restrict AI use in reviews (Naddaf, 2025). Production, in short, scales faster with AI adoption than verification does.

We model this asymmetry in a game-theoretic framework, drawing on strategic models of scientific communication (Zollman et al., 2024; Thurner and Hanel, 2011). We consider a population of authors who choose between two strategies: submitting only genuine work (the cooperative strategy) or submitting everything including low-quality manuscripts (the defecting strategy). A journal applies peer review, subject to both a review capacity K_{rev} and a publication slot constraint K. We introduce an AI adoption intensity parameter $a \in [0, 1]$. Higher a raises submission volume V with elasticity γ, raises review capacity K_{rev} with elasticity $\delta < \gamma$, lowers submission cost c, and raises the false-positive rate λ. The production elasticity exceeds the verification elasticity by assumption, formalizing the asymmetry observed empirically. The net effect of a on the profitability of submitting a low-quality paper, $\Pi_{bad}(x, a) = r \cdot \bar{\rho}(x, a) \cdot \pi_0(x, a) - c(a)$, is positive: AI makes spamming more attractive.

We find three results. First, extreme publication scarcity stabilizes honest submission norms even when screening is imperfect. When the publication lottery is rare enough to win, the expected payoff from submitting a low-quality paper falls below the submission cost, and defection is unprofitable. Scarcity, paradoxically, acts as a deterrent. Second, as a rises, the system can cross a tipping threshold: the long-run outcome shifts abruptly from a high-quality equilibrium to a low-quality one. This tipping occurs because AI simultaneously lowers the break-even point for spam and erodes the screening mechanism that would otherwise deter it. Third, the system exhibits persistent path dependence. The value of a at which quality collapses is strictly lower than the value at which quality recovers. Once the system has tipped into the low-quality long-run outcome, incremental reforms cannot restore it; a large coordinated intervention is needed to cross the recovery threshold.

While expanding publication capacity (raising K) seems like the obvious remedy, we find that doing so without raising review capacity makes the situation worse, not better. More slots increase the expected reward from submitting marginal work, accelerating the drift toward the defecting strategy. Effective intervention must instead target the two primary levers: raising genuine review capacity K_{rev} and reducing the false-positive rate λ. Practical instruments that move these levers include structured code and data audit requirements, submission quotas or tokens, cascading triage systems, and layered certification schemes that decouple preprint dissemination from editorial endorsement. These are not incremental reforms; they constitute a different institutional design, one oriented toward verification rather than volume.

We proceed as follows: Section 4 presents the game-theoretic model; Section **??** derives the replicator dynamics and characterizes the equilibrium regimes; Section 5 examines parameter sensitivity, hysteresis, and strategic levers; and Section 7 concludes.

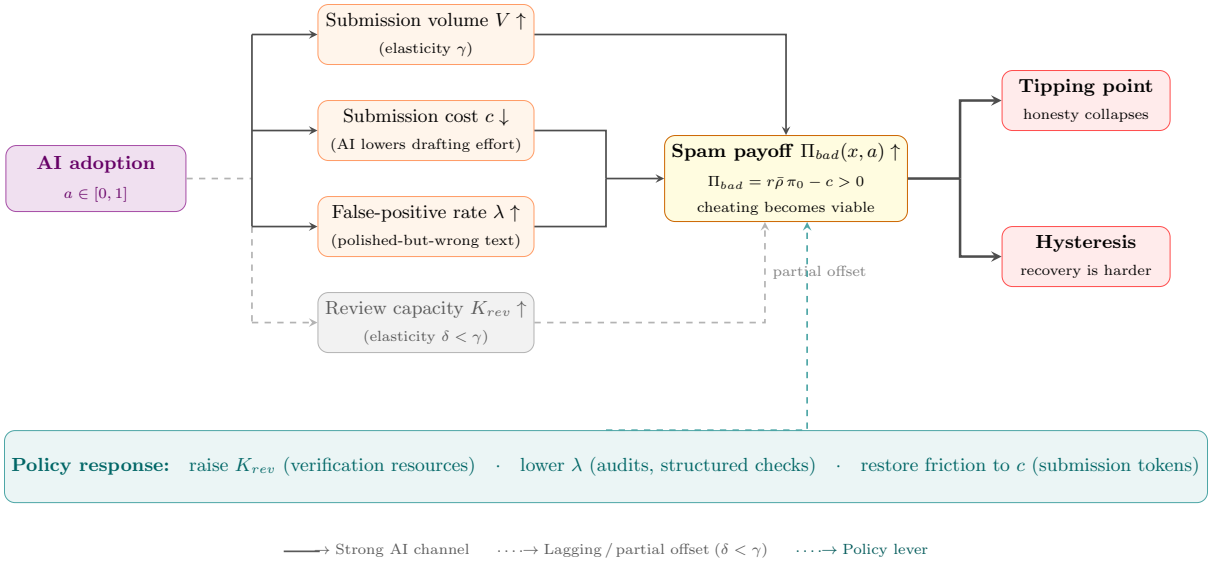


Figure 1: **Mechanism diagram: AI as an asymmetric shock to the publication system.** AI adoption (a) raises submission volume (V, elasticity γ) and lowers drafting cost (c), while polished AI text increases the false-positive rate (λ). Review capacity (K_{rev}) rises more slowly (elasticity $\delta < \gamma$), so the partial offset from AI-assisted reviewing is insufficient to prevent congestion. Together, these shifts raise the profitability of submitting low-quality work ($\Pi_{bad} = r\bar{\rho}\,\pi_0 - c$), pushing the system toward a tipping point and widening the gap between the collapse and recovery thresholds (hysteresis). Effective policy must raise K_{rev}, reduce λ, and restore friction to c (dashed teal arrow).

2 Stylized Facts: AI as an Asymmetric Shock

We draw on evidence that AI dramatically lowers the cost of producing a manuscript. Large language models have cut the time required to draft, revise, and resubmit academic text by a documented margin. In a randomised experiment with 453 college-educated professionals, Noy and Zhang (2023) found that access to ChatGPT reduced the time to complete an incentivised writing task by approximately 40% while simultaneously raising assessed output quality by 18 percentage points, with the largest gains accruing to weaker writers. Stokel-Walker and Van Noorden (2023) report that researchers can revise a full manuscript in under five minutes at an API cost below \$0.50. These micro-level reductions compound at the submission level: iterative resubmission across journals was once a significant labour cost, and AI converts that friction into a near-zero marginal activity, giving every author—regardless of the underlying quality of their work—a cost-effective reason to submit or resubmit. In our model, this channel corresponds to a downward shift in the submission cost parameter $c(a)$.

Aggregate data confirm a sharp rise in submission pressure. Macro-level data are consistent with the micro-level cost story. Monthly submissions to arXiv reached a record 24,226 in October 2024, roughly 50% above the late-2022 baseline; the first month to cross 20,000 submissions was May 2023, shortly after the public release of ChatGPT (arXiv, 2024). More than one-fifth of computer-science preprints sampled in September 2024 showed measurable evidence of large-language-model modification (Liang et al., 2025), and a vocabulary-based analysis of 15 million biomedical abstracts found that at least 13.5% of all 2024 *PubMed* papers contain AI-assisted text, with rates exceeding 40% in some sub-disciplines (Kobak et al., 2025). Alongside this volume growth, annual retraction counts exceeded 10,000 for the first time in 2023, with paper-mill activity implicated in mass retraction events at major publishers. These data show that submission volume V has shifted materially upward as AI adoption has spread, independently of any change in the fraction of authors who genuinely produce high-quality work. In our model, this corresponds to an upward shift in $V(x, a)$ with elasticity γ with respect to a.

While text production scales with AI, verification does not. Rigorous peer review requires reading carefully for logical consistency, checking data against claimed analyses, running or inspecting code, and evaluating genuine novelty—tasks that remain human-intensive in ways that text generation is not. Confidentiality requirements for submitted manuscripts prevent reviewers from uploading papers to third-party AI services, and many journals explicitly forbid AI use in the review process (Naddaf, 2025). Even where AI-assisted reviewing is nominally permitted, the marginal reviewer who can detect a subtle methodological flaw is a scarce specialist whose attention is not fungible with a language model's output. Reviewer acceptance rates have declined continuously for two decades (Meyerson et al., 2025), and the existing burden is already extreme: reviewers contribute more than 100 million hours annually to the peer review system worldwide, with the load concentrated among a small fraction of active researchers (Kovanis et al., 2016). We therefore expect that a system already under strain before the AI era cannot absorb a near-doubling of submission volume through a commensurate increase in review capacity. In our model, this channel is captured by $K_{rev}(a) = K_{rev}(0) \cdot (1 + \delta a)$, where the elasticity δ is strictly smaller than the production elasticity γ, reflecting the structural and policy constraints described above.

AI also polishes manuscripts without improving their reliability, which may raise the false-positive rate. A third channel operates through the quality of the screening signal rather than through volume. AI writing tools produce fluent, well-formatted, and stylistically consistent text (Stokel-Walker and Van Noorden, 2023), and this surface polish reduces the ability of reviewers to distinguish a low-quality submission from a genuine contribution on textual features alone, raising the probability that a flawed manuscript passes the review filter. The screening process degrades further when reviewers themselves turn to AI: between 6.5% and 16.9% of AI-conference peer reviews now contain substantial AI-generated text (Liang et al., 2024), and an experimental audit found that AI-assisted reviews award systematically higher scores, leading to elevated acceptance rates (Latona et al., 2024). Automated detection offers limited recourse—leading AI detectors exhibit false-negative rates above 70%, and adversarial paraphrasing can reduce recall to near zero (Liang et al., 2023). In our model, this channel corresponds to an upward shift in the false-positive rate $\lambda(a)$: as AI adoption rises, the probability that a low-quality submission passes review increases even holding the underlying composition of the author population constant.

We summarize these channels as a single asymmetry hypothesis. The four channels described above—rising submission volume, falling drafting cost, lagging review capacity, and rising false-positive risk—reduce to a single structural claim:

> *AI raises the elasticity of manuscript production above the elasticity of manuscript verification.*

We formalise this as the **AI asymmetry assumption**: with $a \in [0, 1]$ denoting AI adoption intensity,

$$\gamma > \delta \geq 0, \tag{1}$$

where γ is the elasticity of submission volume with respect to a and δ is the elasticity of review capacity with respect to a. This is not an axiom but a testable empirical statement about relative scaling rates. Table 1 translates each channel into a specific model parameter and states the direction of the shift. In Section 5, we examine how our qualitative results depend on the magnitude of the gap $\gamma - \delta$ and on the slope of $\lambda(a)$.

Table 1: **AI adoption channels and their model counterparts.** Each row maps an empirical mechanism to the corresponding model parameter, the direction of the shift, and the evidence base.

Empirical channel	Mechanism	Parameter	Shift	Key evidence
Lower drafting cost	AI reduces time and effort per manuscript; resubmission friction falls	$c(a)$	$\downarrow$	Noy and Zhang (2023); Stokel-Walker and Van Noorden (2023)
Higher submission volume	More authors submit more frequently; low-quality papers enter the pool	V (elasticity γ)	$\uparrow$	arXiv (2024); Liang et al. (2025); Kobak et al. (2025)
Lagging review capacity	Deep verification remains human-intensive; policy restricts AI in review; reviewer shortage pre-dates AI	K_{rev} (elasticity $\delta < \gamma$)	$\uparrow$ slowly	Meyerson et al. (2025); Kovanis et al. (2016); Naddaf (2025)
Rising false-positive rate	Polished AI text harder to distinguish; AI-assisted reviews inflate acceptance scores; detectors unreliable	$\lambda(a)$	$\uparrow$	Liang et al. (2024); Latona et al. (2024); Liang et al. (2023)
Possible quality uplift	AI may also help authors improve genuine contributions (partial offset)	α	$\uparrow$ (offset)	Noy and Zhang (2023)

3 Related Literature

We draw on research showing that peer review functions as a shared resource under increasing strain. Each submission consumes a fixed amount of reviewer time and attention, and when submission volume grows faster than the pool of willing reviewers, that resource degrades. Gross and Bergstrom (2019) show that scientific contests become inefficient at high participation rates: total effort rises, but expected quality per participant falls. Cotton (2013) reaches a parallel conclusion for journals: a small entry cost can deter low-quality submissions and relieve pressure on reviewers. Kovanis et al. (2016) estimate that the global peer review system already consumed over 100 million hours of researcher time each year in the mid-2010s, and Aczél et al. (2021) value the US contribution alone at more than 1.5 billion USD in 2020. Bianchi et al. (2018) model these pressures and find that screening quality declines even when reviewers act in good faith, simply because competing demands leave too little time per paper. The reviewing system, in short, was already near its limits before generative artificial intelligence entered the picture.

While congestion models treat reviewers as passive, a related body of work models authors as strategic agents—and this is where the feedback loop becomes visible. Thurner and Hanel (2011) show that when the review filter is imprecise, submitting everything regardless of quality can become individually rational, pushing the long-run outcome toward low average quality. Zollman et al. (2024) extend this to a multi-journal setting and find that prestige-maximising incentives can reward strategic over honest behaviour. Tiokhin et al. (2021) study honest signalling and show that the good outcome—where only strong work is submitted—is fragile: it collapses when the cost of mimicking quality falls even slightly. Smaldino and McElreath (2016) document this logic empirically, arguing that "publish or perish" incentives act as a selection force that spreads high-volume, low-quality strategies through academic communities. At the aggregate level, Ioannidis (2005) shows that most published findings fail to replicate, not because of fraud, but because structural incentives reward output over reproducibility. Casadevall and Fang (2014) and Stephan (2012) trace this to career and grant structures that treat paper counts as a proxy for scientific value. Paper mills—commercial services producing fabricated manuscripts to order—represent

the extreme end of this logic: annual retraction counts have reached record highs and single incidents have implicated thousands of papers at major publishers. Our model formalises the mechanism underlying these trends: when the review filter is weak and slots are available, submitting low-quality work yields a positive expected payoff, and selection pressure then favours volume over quality.

In contrast to the strategic submission literature, recent empirical work documents how generative artificial intelligence reshapes both sides of the system simultaneously. On the production side, Noy and Zhang (2023) find that access to a large language model cuts professional writing time by 40% and raises rated output quality by 18 percentage points. Liang et al. (2025) detect language-model modification in more than one-fifth of computer-science preprints by late 2024, and Kobak et al. (2025) find AI-assisted text in at least 13.5% of biomedical abstracts from the same period. On the evaluation side, however, the picture is more troubling. Checco et al. (2021) conclude that deep verification—checking data, code, and logical consistency—remains beyond current automated systems. Liang et al. (2024) find that between 6.5% and 16.9% of reviews at major conferences already contain AI-generated text, and Latona et al. (2024) show experimentally that AI-assisted reviews award higher scores, raising acceptance rates. Naddaf (2025) reports that more than half of surveyed researchers now use AI tools for reviewing, often against journal policy. These findings supply the empirical foundation for our asymmetry assumption: artificial intelligence raises submission productivity faster than it raises verification capacity.

A fourth strand addresses the economics of gatekeeping and institutional design. Ellison (2002) shows that the equilibrium publication standard depends on how strongly readers value quality signals. Muller-Itten (2022) finds that journals maintain strict standards to protect their certification value: accepting too freely devalues their own signal. Casadevall and Fang (2014) argue that effective reform must target the false-positive rate directly—through data deposition, structured reporting, and replication requirements—rather than adjusting slot counts alone. Ginsparg (2011) shows that dissemination and certification can be separated: preprint servers make manuscripts available immediately while editorial endorsement follows independently. Cotton (2013) provides the sharpest economic case for submission fees: a small cost of entry restores friction, filters marginal submissions, and reduces reviewer overload.

What we contribute

While these literatures each address part of the problem, none unifies them. Models of strategic submission do not treat artificial intelligence as a structural shock that shifts multiple inputs at once. The empirical AI-in-science literature documents faster production, slower review growth, and higher false-positive risk, but does not connect these to a model with clear predictions about the long-run state of the system. Neither strand addresses the possibility that submission pressure can grow gradually until it crosses a threshold, after which the system shifts suddenly to a low-quality state and resists recovery. The institutional design literature does not ask how its proposals must change when drafting cost approaches zero and the false-positive rate rises continuously with adoption. We close these gaps by connecting all four channels—$c(a)$ falling, V rising, K_{rev} lagging, and $\lambda(a)$ rising—to the long-run state of the publishing system, characterising conditions under which quality collapses suddenly and is hard to restore, and deriving which institutional levers can prevent or reverse that shift.

4 Baseline Model

Players and strategies. We consider a population of N authors submitting to a single journal over repeated periods. Each manuscript has binary quality: good ($q = 1$) with probability α, and bad ($q = 0$) with probability $1 - \alpha$. Standard dominance arguments reduce the strategy space to two options (Fudenberg and Tirole, 1991). Under **selective submission** (OG), an author submits only when $q = 1$. Under **universal submission** (AS), an author submits regardless of quality, hoping that imperfect screening allows bad papers through. We write $x \in [0, 1]$ for the share of authors playing OG; the remaining $1 - x$ play AS.

Two bottlenecks. The publication process has two distinct chokepoints; each creates a different strategic force. The first is **review capacity** K_{rev}: the number of papers that can receive careful scrutiny

per period. Submission volume is $V(x) = N[1 - (1 - \alpha)x]$, which rises as the AS share grows. When V exceeds K_{rev}, only a fraction of submissions are carefully reviewed; the rest pass without scrutiny—a rubber-stamping effect documented in overloaded systems (Azar, 2005; Kovanis et al., 2016). The **screening intensity** is:

$$\eta(x) = \min\left(1,\ \frac{K_{rev}}{V(x)}\right). \tag{2}$$

The second bottleneck is **publication capacity** K: the number of slots the journal can fill per period. Papers that pass review form a peer-accepted pool $M(x)$; at most K are published. The probability that a peer-accepted paper is ultimately published—the **rationing probability**—is:

$$\bar{\rho}(x) = \min\left(1,\ \frac{K}{M(x)}\right). \tag{3}$$

Screening errors and the spam payoff. Peer review is an imperfect filter (Bornmann et al., 2010; Cicchetti, 1991). A good paper is falsely rejected with probability ϵ; a bad paper wrongly passes with probability λ. Both error rates apply only to papers that are actually reviewed. When $\eta < 1$, unscreened papers pass automatically, raising the effective pass rates:

$$\pi_0(x) = 1 - \eta(x)(1 - \lambda), \qquad \pi_1(x) = 1 - \eta(x)\,\epsilon. \tag{4}$$

As $\eta \to 0$, both rates converge to 1: the journal loses all discriminatory power. The peer-accepted pool is $M(x) = N\alpha\,\pi_1(x) + N(1 - \alpha)(1 - x)\,\pi_0(x)$. Because OG and AS authors submit good papers on identical terms, the strategic difference reduces to the net value of submitting a bad paper:

$$\Pi_{bad}(x) = r \cdot \bar{\rho}(x) \cdot \pi_0(x) - c, \tag{5}$$

where r is the publication reward and c the submission cost. We work in the region $\lambda r < c < r(1 - \epsilon)$: without congestion, submitting a bad paper is unprofitable, but the incentive reactivates once screening degrades—the same adverse-selection logic as in Akerlof's market for lemons (Akerlof, 1970).

Dynamics. Strategy shares evolve according to the replicator equation (Weibull, 1995; Hofbauer and Sigmund, 1998):

$$\dot{x} = -x(1 - x)(1 - \alpha)\,\Pi_{bad}(x). \tag{6}$$

When $\Pi_{bad} > 0$, AS spreads and x falls; when $\Pi_{bad} < 0$, OG spreads and x rises. Two forces compete as the AS share grows. First, rising volume degrades screening (η falls, π_0 rises): submitting bad papers becomes more attractive. Second, rising volume floods the publication queue ($\bar{\rho}$ falls): the publication lottery becomes harder to win. When the screening-collapse force dominates, the system tips toward full corruption. When the rationing force dominates, extreme slot scarcity deters spam even after screening has failed—a mechanism we call *defence by congestion*.

Four long-run outcomes. The signs of Π_{bad} at the two boundary states determine which long-run outcome the system reaches. Table 2 summarises the four cases. A key structural observation is that the two boundaries are controlled by different parameters: the honest boundary ($x = 1$) is secured by keeping λ low, while the corrupt boundary ($x = 0$) is held by keeping K tight. This means the two bottlenecks can be targeted separately as policy instruments—a point we return to in Section 5. Figure 2 illustrates the bistable case. Π_{bad} is positive near $x = 0$ (corruption reinforces itself) and negative near $x = 1$ (honesty reinforces itself), with an unstable tipping threshold at $x^* \approx 0.82$ separating the two regions. Which long-run outcome prevails depends entirely on where the system starts relative to this threshold.

Table 2: **Four long-run outcomes determined by boundary payoffs.** $\Pi_{bad}(1)$ is the spam payoff when almost all authors are honest (low-volume limit); $\Pi_{bad}(0)$ is the spam payoff when almost no authors are honest (maximum-load limit).

Regime	$\mathbf{\Pi_{bad}(1)}$	$\mathbf{\Pi_{bad}(0)}$	**Long-run outcome**
GH	< 0	< 0	**Global honesty.** Spamming is unprofitable everywhere. The system reaches $x = 1$ from any starting point. Primary lever: keep $\lambda < c/r$.
GC	> 0	> 0	**Global corruption.** Spamming is profitable everywhere. The system collapses to $x = 0$ from any starting point. Seen in predatory journals with weak review and ample slots.
BS	< 0	> 0	**Bistability.** Both $x = 1$ and $x = 0$ are long-run outcomes. An unstable tipping threshold x^* separates the regions; history determines which outcome prevails. Recovery from corruption requires a large intervention.
SC	> 0	< 0	**Stable coexistence.** Neither extreme is reached. A stable interior outcome x^* emerges, sustained by rationing. Seen in top-tier journals where the publication lottery is very hard to win.

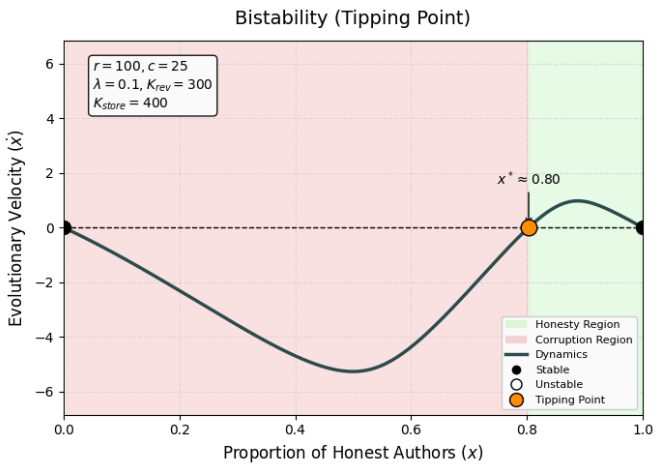


Figure 2: **Baseline phase portrait (bistability).** The curve plots $\Pi_{bad}(x)$, the net payoff from submitting a bad paper, against the honest-author share x. Parameters: $N = 1000$, $K_{rev} = 300$, $K = 400$, $r = 100$, $c = 25$, $\lambda = 0.1$. Red shading: $\Pi_{bad} > 0$, so AS spreads and x falls. Green shading: $\Pi_{bad} < 0$, so AS shrinks and x rises. Two long-run outcomes coexist—full honesty ($x = 1$) and full corruption ($x = 0$)—separated by an unstable tipping threshold at $x^* \approx 0.82$ (orange marker). Which outcome prevails depends entirely on the starting position relative to x^*.

5 Parameter Sensitivity and Regime Transitions

5.1 Methodology and Baseline Calibration

We want to know which policies maximize publication quality and minimize academic misconduct. However, due to the non-linear interplay between the review bottleneck (η) and the storage bottleneck ($\bar{\rho}$), the system does not respond smoothly to policy changes. Instead, the ecosystem undergoes phase transitions between qualitatively distinct regimes.

To analyze these transitions, we use numerical bifurcation analysis. We define the instantaneous Average Publication Quality $Q(x)$ as the proportion of good papers within the set of accepted publications, a metric that aligns with standard meta-scientific frameworks for assessing the reliability of research findings (Ioannidis, 2005):

$$Q(x) = \frac{N\alpha \cdot \pi_1(x)}{M(x)} = \frac{\alpha\pi_1(x)}{\alpha\pi_1(x) + (1-\alpha)(1-x)\pi_0(x)} \tag{7}$$

Crucially, the long-term quality Q_∞ is determined by the system's stable attractors. In bistable regimes, Q_∞ is path-dependent, meaning the final quality depends on whether the system started from a high-trust or low-trust state.

We classify the system into four regimes based on the stability of equilibria:

1. **Global Honesty (GH):** the system converges to $x = 1$, yielding $Q_\infty = 1.0$.
2. **Global Corruption (GC):** the system collapses to $x = 0$, yielding $Q_\infty \approx \alpha$—a "market for lemons" where the journal fails to filter noise (Akerlof, 1970).
3. **Bistability (BS):** two stable attractors coexist and the outcome depends on history.
4. **Stable Coexistence (SC):** an interior equilibrium x^* is maintained by congestion, stabilizing quality at a mediocre level ($Q_\infty \approx 0.5$).

To ensure comparative consistency, we use two representative parameter sets (Table 3). Set A represents a typical journal with moderate capacity, prone to bistability. Set B represents a top-tier journal with strict page limits and high rewards, prone to stable coexistence.

Table 3: Parameter Calibration for Numerical Analysis

Symbol	Description	Set A (Baseline)	Set B (High-Pressure)
N	Total Population	1000	1000
α	Author Productivity	0.2	0.2
K_{rev}	Review Capacity	**300**	**1200**
K	Slot Capacity	**400**	**180**
r	Reward	100	**150**
c	Cost	25	**30**
λ	False Positive Rate	0.1	**0.4**
ϵ	False Negative Rate	0.05	0.01

Note: Set A is used as the default for sensitivity sweeps. Set B is used to illustrate the defense-by-congestion mechanism.

Why two parameter sets? A single baseline cannot capture the full range of the system's dynamical behaviours, because the dual-bottleneck mechanism is non-monotonic. We distinguish two archetypes. Set A represents the default state of many modern journals: moderate prestige ($r = 100$) and limited but not extreme storage constraints ($K = 400$). Here the primary failure mode is the review bottleneck ($K_{rev} = 300$)—when submission volume rises, the review process collapses before the storage limit is

reached. This setup reflects the empirical reality of the "global burden" of peer review, where reviewer scarcity is the dominant constraint (Kovanis et al., 2016), and it generates a bistable topology in which the system is fragile and prone to catastrophic tipping. Set B represents top-tier prestige journals such as *Nature* or *Science*, where the reward is significantly higher ($r = 150$) and the defining constraint is not the lack of reviewers (we assume ample review capacity $K_{rev} = 1200$) but the extreme scarcity of publication slots ($K = 180$). This aligns with empirical findings on scientific gatekeeping, where high rejection rates are driven by physical slot limits rather than quality filtering alone (Siler et al., 2015; Casadevall and Fang, 2014), and it generates the stable coexistence topology that illustrates what we call the defence-by-congestion mechanism. By analysing both sets, we decouple the effects of review collapse from slot rationing, ensuring that our conclusions are robust across different segments of the academic publishing hierarchy.

5.2 The Decoupling of Capacities: Phase Diagram Analysis

Standard academic models often treat "journal capacity" as a single resource. Our dual-bottleneck framework reveals that the system's fate depends on how two distinct capacities interact: review capacity K_{rev}, which determines the screening intensity $\eta(x)$, and slot capacity K_{store}, which determines the rationing probability $\bar{\rho}(x)$. High K_{rev} prevents review collapse; low K_{store} creates what we call a defence by congestion.

To visualize the interaction between these forces, we perform a two-dimensional bifurcation scan across the parameter space (K_{rev}, K_{store}). The resulting phase diagram (Figure 3) maps the system's regimes.

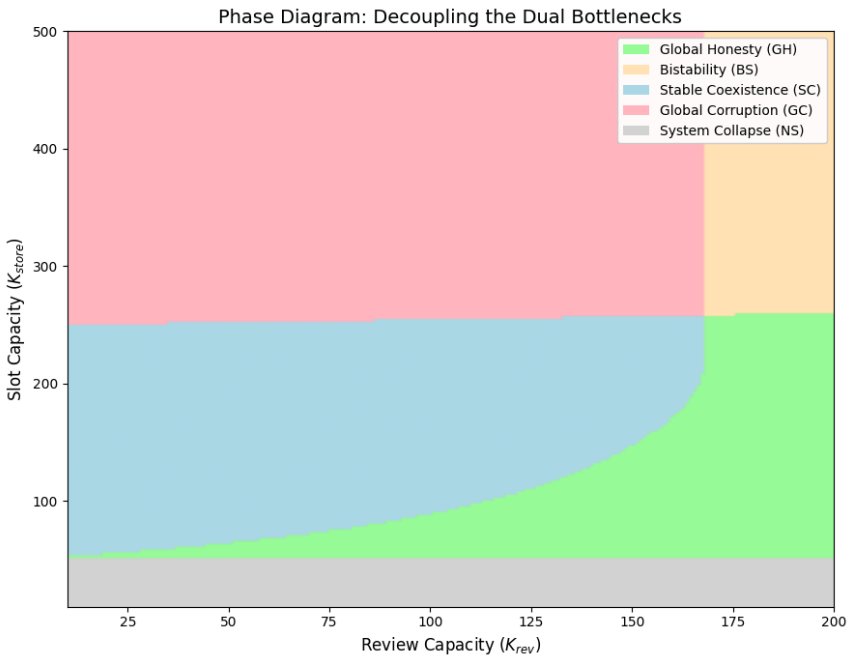


Figure 3: **Phase Diagram of the Dual-Bottleneck System.** The map classifies the topological regime based on the availability of review resources (K_{rev}) versus publication slots (K_{store}). **Green (Global Honesty):** High review capacity ensures cheating is unprofitable ($Q \to 1.0$). **Blue (Stable Coexistence):** Extreme slot scarcity ($K_{store} \ll N$) triggers the "Defense by Congestion," stabilizing the system at a mediocre but functional level. **Yellow (Bistability):** A transition zone where the outcome is history-dependent. **Red (Global Corruption):** When slots are abundant but review is weak, the defense mechanism vanishes, leading to total collapse ($Q \to \alpha$). **Gray (System Collapse):** Feasibility constraint violated (K_{store} too low); even honest authors cannot break even.

The topology of Figure 3 highlights two critical dynamical phenomena:

The paradox of slot expansion. Intuition suggests that giving a journal more publication slots should improve the system. We find the opposite. Moving vertically from the bottom of the diagram corresponds to increasing K_{store}. Initially, increasing slots rescues the system from total collapse (Gray to Blue), making participation rational for honest authors. However, further increasing K_{store} pushes the system from stable coexistence (Blue) into global corruption (Red) or bistability (Yellow). This paradox arises because tight storage constraints act as a passive defence by congestion, keeping $\bar{\rho}$ low enough to deter spam. By increasing K_{store}, the journal inadvertently dismantles this defence: $\bar{\rho}$ rises, the publication lottery becomes winnable again, and if K_{rev} is not increased simultaneously, the incentive to cheat reactivates and the system tips into full corruption. This finding aligns with strategic gatekeeping models showing that expanding capacity without scaling review rigour destroys the signal value of the journal (Muller-Itten, 2022).

The necessity of review resources. Moving horizontally to the right corresponds to investing in review capacity K_{rev}. Provided the system is in a feasible region (not Gray), we find that increasing K_{rev} is unconditionally beneficial, expanding the Global Honesty (Green) zone. Crucially, the boundary between corruption and honesty slopes upward: as K_{store} increases, the journal requires *disproportionately* more review capacity to maintain honesty. This implies that larger journals are inherently harder to police than smaller ones—they require a review infrastructure that scales super-linearly with volume to maintain the same level of integrity, exacerbating the already critical burden of peer review (Kovanis et al., 2016).

5.3 Population Pressure and Hysteresis: The Cost of Lost Trust

While capacity parameters define the structural limits of the journal, the total population N represents the external load. One might expect quality to decline smoothly as load increases. We find instead that the system exhibits hysteresis in the bistable regime: the value of N at which quality collapses is much higher than the value at which it recovers. This is consistent with theories of catastrophic shifts in complex systems, where state recovery requires significantly more effort than state maintenance (Scheffer et al., 2001).

To capture this path dependence, we use a numerical adiabatic continuation method. We define the system state $x_{st}(N)$ as the stable equilibrium selected by continuity from the previous state:

$$x_{st}(N_i) = \arg\min_{x \in \mathcal{E}_{st}(N_i)} |x - x_{st}(N_{i-1})| \tag{8}$$

where $\mathcal{E}_{st}(N_i)$ is the set of stable fixed points at N_i.

- **Forward Path (Crowding):** Increments N, initializing from $x_{st}(N_{min}) \approx 1$.
- **Backward Path (Recovery):** Decrements N, initializing from $x_{st}(N_{max}) \approx 0$.

Figure 4 illustrates the resulting trajectories using Baseline Set A.

We highlight three features of the result.

Structural resilience (forward path). A journal with a strong reputation demonstrates remarkable resilience. As seen in the blue trajectory, the system maintains $x_{st} \approx 1$ and perfect quality ($Q = 1.0$) even when N far exceeds the nominal capacity, because honest authors self-regulate, keeping the effective load $V(1) = N\alpha$ manageable. The collapse point $N_\uparrow \approx 1800$ is the threshold where the honest equilibrium loses stability: $N_\uparrow = \inf\{N : \Pi_{bad}(1; N) \geq 0\}$.

The stickiness of corruption (backward path). Once the system settles into the corrupt equilibrium, the sheer volume of spam saturates the review capacity. Even if N is reduced, the system remains trapped in the low-quality state as long as $\Pi_{bad}(0; N) > 0$. The recovery point $N_\downarrow \approx 350$ corresponds to the threshold where the corrupt equilibrium finally becomes unstable. This asymmetry mirrors the

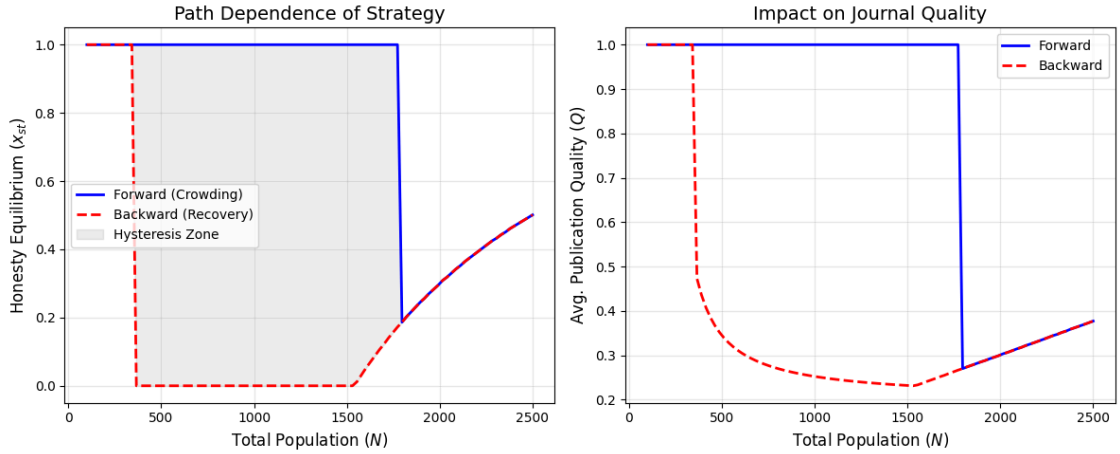


Figure 4: **Hysteresis Loop and Quality Collapse (Baseline Set A).** The shaded region represents the bistable interval $[N_\downarrow, N_\uparrow]$. **Left Panel (State):** On the Forward Path (Blue), the "Honest Norm" is resilient ($x_{st} \approx 1$) until the collapse threshold $N_\uparrow \approx 1800$. Once corrupted (Red Dashed), the system is trapped in the low-honesty attractor ($x_{st} \approx 0$). Recovery requires reducing N below the distinct recovery threshold $N_\downarrow \approx 350$. **Right Panel (Quality):** The collapse leads to a discrete jump in quality. In the corrupt state (Red Dashed), quality is dominated by the baseline productivity ($\alpha = 0.2$) but fluctuates slightly as review intensity $\eta(N)$ varies. Recovery only occurs when the system crosses $N_\downarrow$.

dynamics of collective reputation, where a group remains locked in a low-quality equilibrium long after external conditions improve, because the burden of proof is too high for individual reformers (Tirole, 1996).

Crucially, we define the Average Publication Quality $Q(x)$ as the proportion of good papers in the passed pool $M(x)$:

$$Q(x) = \frac{N\alpha\pi_1(x)}{N\alpha\pi_1(x) + N(1-\alpha)(1-x)\pi_0(x)} \tag{9}$$

Since slot rationing ($\bar{\rho}$) applies uniformly to $M(x)$, it does not alter the ratio of good-to-bad papers. Consequently, quality remains low throughout the backward path until the critical transition occurs.

Policy implication: basin hopping. The disparity between $N_\uparrow$ and $N_\downarrow$ implies that incremental reforms are structurally insufficient for a corrupted journal. Within the bistable interval, marginal reductions in submissions do not change the long-run attractor. Escaping the low-quality basin requires interventions strong enough to move the system across the recovery threshold ($N < N_\downarrow$). This suggests that while soft measures work for maintenance, hard measures (e.g., temporary submission moratoriums) are structural necessities for recovery.

5.4 Strategic Levers and Regime Transitions

While capacity parameters define the structural limits, the strategic parameters $(r, c, \lambda, \epsilon, \alpha)$ determine the incentives. We perform a comparative sensitivity analysis across all five dimensions, grouping them into three functional categories: screening, economic incentives, and productivity.

Screening parameters: the false-positive rate versus friction. We begin with the two parameters controlling the peer review filter (Figure 5). Panel (a) reveals that the false-positive rate λ is the most critical parameter in the entire system. As λ increases, the system shifts rapidly from Green (Honesty) to Yellow (Bistability) and finally to Red (Corruption); in the yellow zone, a minor perturbation can trigger a catastrophic tip into the corrupt equilibrium. This aligns with signalling theory, where the reliability of the signal (low λ) is paramount for maintaining the separating equilibrium (Tiokhin et al., 2021).

In contrast, Panel (b) shows that friction ϵ has negligible structural impact: within the explored range, the regime boundaries remain largely unchanged. This implies that policies focused solely on reducing administrative friction for good papers yield little improvement in integrity compared to improving detection accuracy.

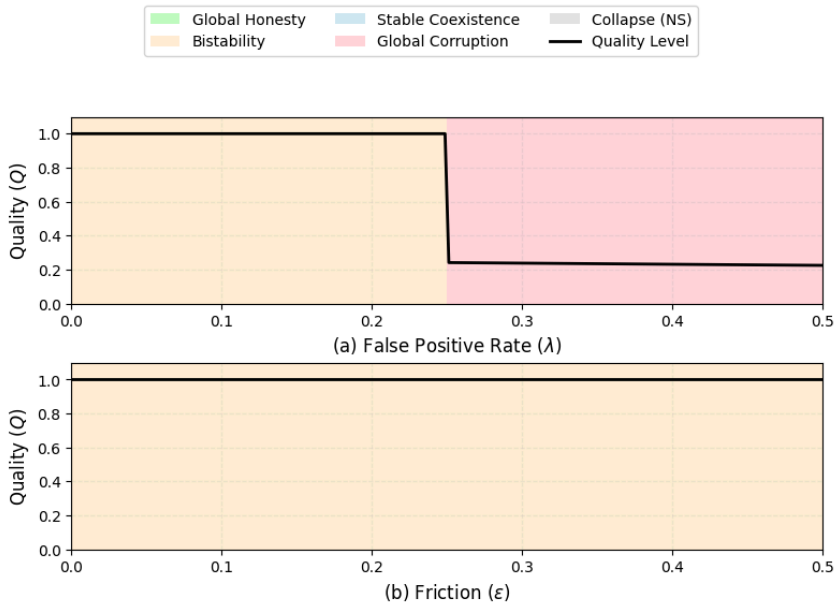


Figure 5: **Screening Parameters and Regime Stability. (a) False Positive Rate λ:** Acts as a major destabilizer. Increasing λ destroys the honest equilibrium, pushing the system into high-risk bistability. **(b) Friction ϵ:** Has negligible impact on regime boundaries, acting merely as a silent administrative cost.

Economic levers: reward and submission cost. We next examine the financial and reputational incentives (Figure 6). Panel (c) demonstrates that high rewards r act as an accelerant for corruption, significantly expanding the Red and Yellow zones. When publication rewards are excessive, the lottery of rubber-stamping becomes profitable even at low probabilities—confirming the mechanism of "natural selection of bad science," where high incentives select for strategies that maximize output at the expense

of quality (Smaldino and McElreath, 2016).

While reward drives the system toward corruption, submission cost c pulls it back. Panel (d) shows that increasing c effectively shifts the system from Red to Green, providing theoretical support for submission fees as a tool to reduce congestion and filter out low-quality submissions (Cotton, 2013). However, pushing costs too high forces the system into the Gray zone, where even honest authors cannot sustain participation ($\Pi_{good} < 0$).

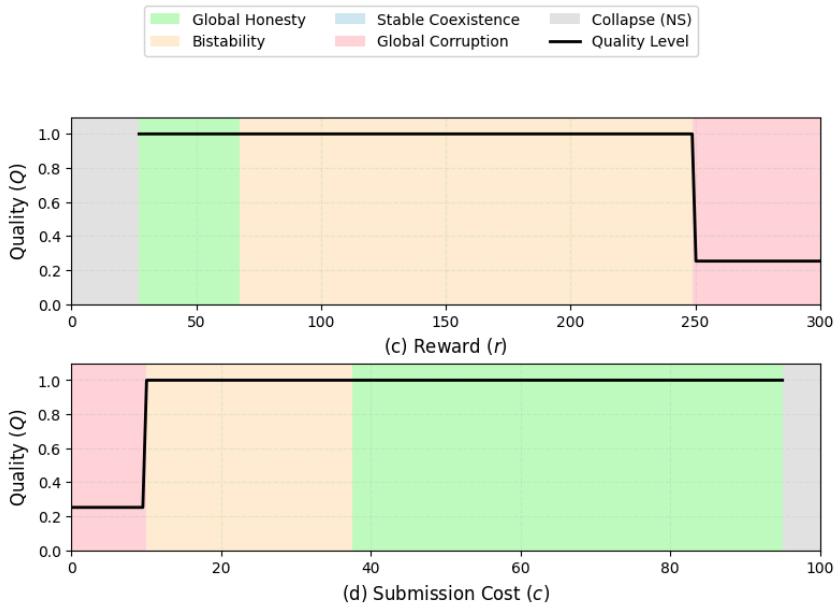


Figure 6: **Economic Incentives and Filtering. (c) Reward r:** High rewards expand the corrupt zones (Red/Yellow), incentivizing spamming strategies. **(d) Cost c:** Submission fees filter out spam, restoring Green, but excessive costs kill the journal (Gray).

The productivity paradox. Finally, we analyse the impact of aggregate author productivity (Figure 7). One might expect that higher productivity α would stabilize the system, since the input mix improves. We find the opposite: as α increases, the background topology shifts toward unstable regimes (Yellow/Red), even though the average quality (black line) rises mechanically. The mechanism is congestion. As legitimate authors produce more papers, the total volume rises and consumes scarce review capacity; if K_{rev} is fixed, this "good congestion" dilutes the screening intensity for everyone, paradoxically making the system more vulnerable to invasion by spammers.

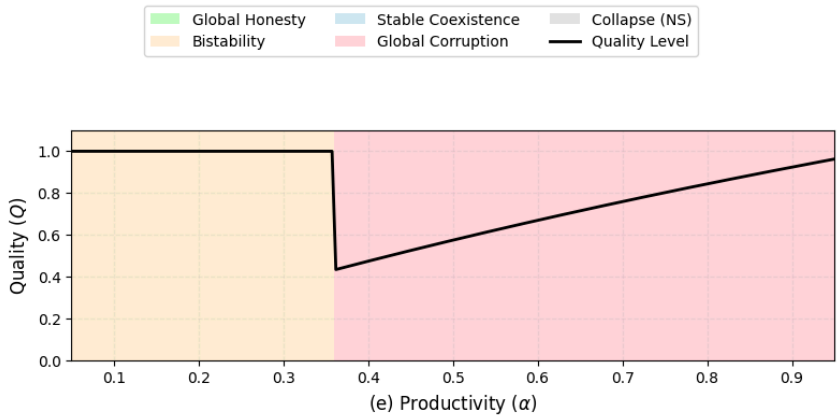


Figure 7: **The Productivity Paradox. (e) Productivity α:** Higher productivity increases total congestion. While observable quality rises (black line), the structural stability degrades (background shifts to Red), narrowing the window of stability.

6 Agent-Based Modeling

6.1 Simulation Design

The analytical results above assume a large, well-mixed population and deterministic strategy evolution—the standard mean-field approximation (Hofbauer and Sigmund, 1998). Real academic communities are finite and subject to stochastic fluctuations (Nowak, 2006). We therefore implement an agent-based model (ABM) to test whether our main claims survive these complications: that dominated strategies die out naturally, that the predicted long-run outcomes emerge from local interactions under noise, and that the defence-by-congestion mechanism is not an artefact of the mean-field limit (Bonabeau, 2002).

We simulate a population of $N = 1000$ agents, following the standard protocols of stochastic evolutionary game theory in finite populations Traulsen and Hauert (2009). We initialise the population with a uniform distribution across all four strategies (25% each), so that the simulation must find the reduced state on its own rather than having it assumed. The process proceeds as follows:

1. **Interaction Phase:** In each time step t, agents generate submissions based on their full strategy space ($s_i \in \{\text{OG}, \text{AS}, \text{OB}, \text{NS}\}$). The aggregate volume V and peer-accepted volume M are calculated endogenously, determining the congestion levels η and ρ.

2. **Payoff Realization:** Agents receive payoffs Π_i based on the realized publication outcomes. This captures the "lottery" nature of the congestion bottleneck.

3. **Strategy Update (Social Learning):** We employ the **pairwise comparison process** to model strategy evolution. A randomly selected agent A considers mimicking agent B's strategy. The

probability of switching $s_A \to s_B$ is given by the Fermi function derived from statistical mechanics Szabó and Tőke (1998):

$$P(s_A \to s_B) = \frac{1}{1 + \exp(-\beta[\Pi_B - \Pi_A])} \tag{10}$$

where β represents the intensity of selection. This update rule is the standard method for introducing bounded rationality and decision noise into evolutionary dynamics Traulsen and Hauert (2009).

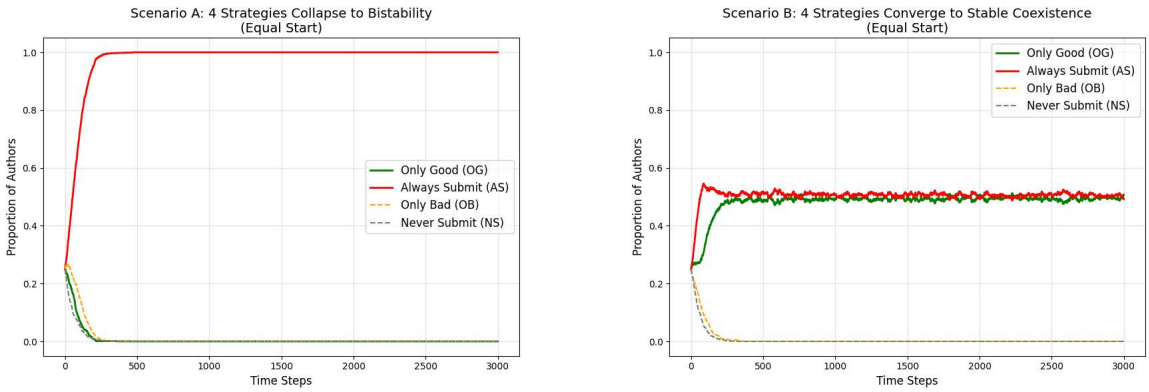
6.2 Simulation Results

We simulated both parameter sets from Table 3 with a high selection intensity ($\beta = 1.0$) to reduce noise. Figure 8 shows that the agent-based simulations corroborate both the strategy elimination and the macroscopic predictions from Section 3.

In both scenarios, the "Only Bad" (OB) and "Never Submit" (NS) strategies are rapidly driven to extinction within the first few hundred time steps, directly verifying the iterated elimination of strictly dominated strategies (IESDS) conducted in Section 2.6. The system autonomously reduces to a binary contest between OG and AS, consistent with the finding that strictly dominated strategies cannot survive under monotonic selection dynamics Samuelson (1997).

In the review-dominated regime (Scenario A), the dynamics exhibit sharp bistability (Figure 8a). Because the initial proportion of honest OG authors (25%) falls well below the theoretical unstable tipping point ($x^* \approx 0.82$), the system cannot self-correct. Once OB and NS vanish, the AS strategy outcompetes the remaining honest authors, driving the system into the absorbing state of full corruption ($x = 0$). This confirms the severe path dependence predicted by the analytical model Scheffer et al. (2001); Schelling (1978).

In the rationing-dominated regime (Scenario B), we observe a qualitatively different outcome (Figure 8b). Despite the initial chaotic interaction of all four strategies, the population rapidly settles to a long-run average of $x^* \approx 0.49$ for OG and AS, while OB and NS remain extinct. This result aligns precisely with the analytical fixed point ($x^* \approx 0.49$) identified in the phase portrait analysis, confirming that the defence-by-congestion mechanism is a robust emergent property, immune to initial strategy turbulence and finite-size stochasticity.



(a) Scenario A: Review-Dominated Dynamics

(b) Scenario B: Stable Coexistence

Figure 8: **Agent-Based Model Trajectories with 4 Strategies ($\beta = 1.0$).** The plots show the time evolution of the strategy proportions. **(a)** Strong Bistability: Starting from an equal uniform mix, the system rapidly eliminates OB and NS. Because the initial OG fraction (0.25) is below the critical tipping point, the system collapses to full corruption ($x = 0$). **(b)** Stable Coexistence: OB and NS are quickly eliminated, and the remaining OG and AS populations converge to a mixed equilibrium ($x^* \approx 0.49$). The precise agreement with analytical predictions confirms both the theoretical dominance analysis and the structural congestion stabilization.

6.3 Co-operation ABM

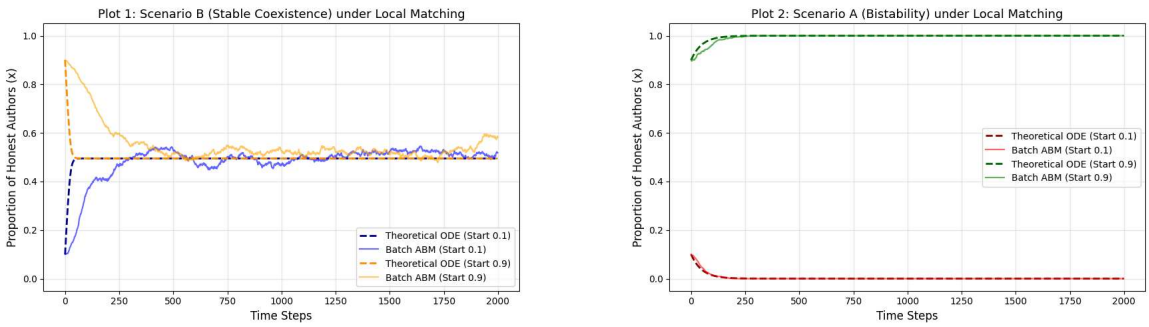
To rigorously validate the robustness of our theoretical findings and explore the microscopic foundations of the "Defense by Congestion" mechanism, we develop a specialized Co-operation Agent-Based Model (ABM) featuring local batch-matching. While the standard replicator dynamics rely on a mean-field approximation assuming an infinite population and a single monolithic market, real-world academic publishing is inherently fragmented. Authors typically compete within specific sub-disciplines, specialized journals, or distinct temporal submission windows. The primary motivation for designing this ABM is to investigate whether the macroscopic stability predicted by the ODEs holds when subjected to the localized uncertainty, heterogeneous payoffs, and stochastic fluctuations characteristic of finite populations.

The core innovation of this extended ABM is the *batch-matching* algorithm. Instead of evaluating all N agents globally, the population is randomly partitioned into local batches (sub-markets) in each time step. The journal's review and storage capacities (K_{rev} and K_{store}) are scaled proportionally to match these local batches. This approach significantly improves upon standard global ABMs by introducing critical variance into the localized rationing probability (ρ) and enabling stochastic drift. It captures the granular reality that the publication "lottery" is not uniform; different sub-fields experience varying degrees of congestion simultaneously. This localized competition provides a much more rigorous stress-test for the theoretical equilibria.

The simulation results validate our threshold competition framework across two key dimensions, as

illustrated in Figure 9:

Trajectory Analysis and Robustness: Figures 9(a) and 9(b) illustrate the time evolution of strategies under batch matching. In the rationing-dominated regime (Scenario B), despite chaotic local interactions, both the theoretical ODE and the ABM trajectories converge tightly to the stable internal equilibrium of $x^* \approx 0.49$. This confirms that the stable coexistence induced by congestion is highly robust against local noise. Conversely, in the review-dominated regime (Scenario A), the ABM demonstrates sharp bistability, collapsing rapidly to global honesty ($x = 1.0$) or global corruption ($x = 0.0$) depending on initial conditions. The ABM closely traces the expected theoretical paths but exhibits micro-oscillations, validating the IESDS reduction while underscoring the severe path-dependence of the bistable regime.



(a) Scenario B (Stable Coexistence) Trajectories

(b) Scenario A (Bistability) Trajectories

Figure 9: **Results of the Co-operation ABM with Local Batch-Matching.** The composite figure demonstrates that the theoretical mean-field predictions hold even under the rigorous localized constraints and stochastic noise introduced by fragmented sub-markets.

The Journal as an Independent Evolutionary Agent

From Fixed Environment to Co-Evolutionary Agent In previous sections, including the local batch-matching analysis in Section 5.4, the journal was mathematically treated as a fixed institutional player locked into the "Only Good" (OG) strategy. It acted purely as a static environmental constraint enforcing a peer review filter and a capacity limit. By elevating the journal to an independent, adaptive agent, we transition the framework from a simple population game into an asymmetric co-evolutionary game. Journals now possess their own complete strategy space, consisting of Only Good (OG), Always Accept (AA), Only Bad (OB), and Reject All (AR). They update their behavior based on their own distinct utility function, which must balance the reputation benefits of publishing good papers, the penalties of publishing bad papers, and the administrative costs of peer review. This structural update captures the "Red Queen" dynamic inherent in academic publishing: if the author population becomes overwhelmingly honest, journals are financially incentivized to drop their costly screening processes and adopt the "Always Accept" (AA) strategy. However, this relaxation immediately triggers an oppor-

tunistic invasion of "Always Submit" (AS) authors, which in turn forces the journals to revert to strict filtering.

The Significance of the 10:1 Local Batch-Matching Mechanism Real-world academic publishing is not a monolithic global market, but rather a highly fragmented ecosystem divided into specific sub-disciplines and specialized journals. To accurately model this micro-level reality, we introduce an extreme 10:1 local batch-matching mechanism, pairing 1000 authors with 100 distinct journal agents. In this configuration, global capacities (K_{rev} and K_{store}) are scaled down proportionally to match the size of these local sub-markets. The primary significance of this 10:1 ratio is the maximization of localized variance and stochastic drift. By breaking the assumption of uniform global congestion, a single journal agent shifting to a rubber-stamping (AA) strategy creates a massive, localized payoff spike for the specific batch of 10 authors assigned to it, severely testing the structural resilience of the system.

ABM Configuration and Result Analysis To simulate this environment, we employ an Agent-Based Model (ABM) where both populations update their strategies via a Fermi pairwise comparison process, introducing bounded rationality and decision noise.

First, we analyze the global co-evolutionary baseline without fragmentation, as illustrated in **Figure 10**. The ABM realization demonstrates that structurally dominated strategies, such as Only Bad (OB) and Reject All (AR), are rapidly driven to extinction. The remaining strategies enter a dynamic coexistence where the continuous mathematical predictions (ODE) and the finite-population ABM trajectories align closely.

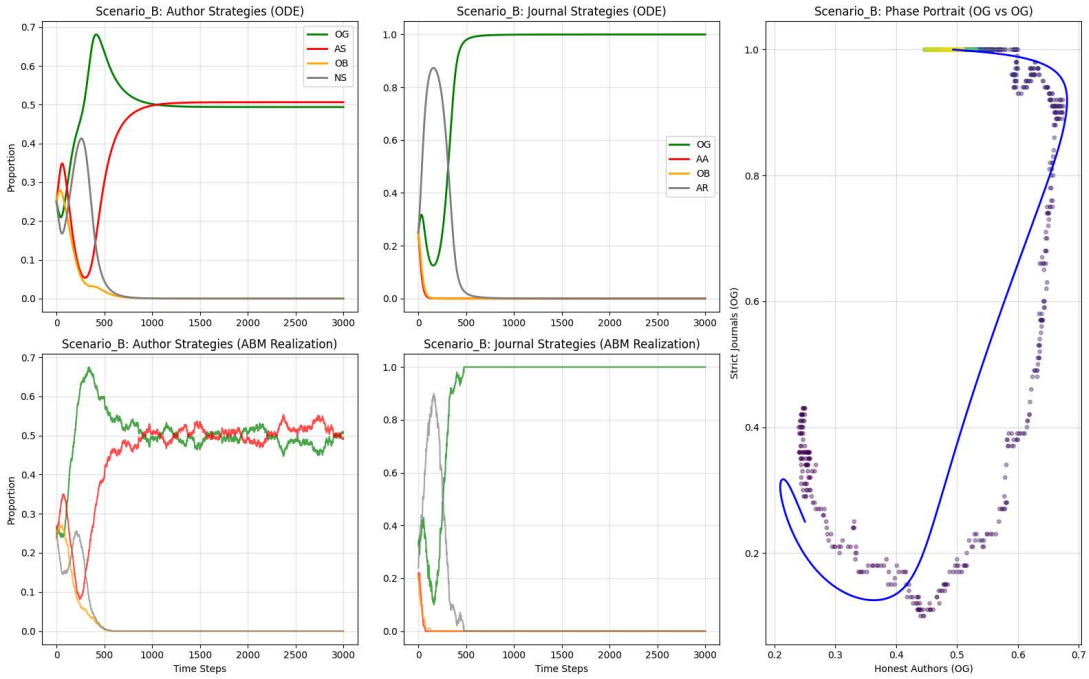


Figure 10: Global Co-evolutionary Baseline (Scenario B). The system demonstrates stable coexistence without fragmentation, where the finite-population ABM trajectory closely aligns with the continuous ODE predictions.

However, under the extreme fragmentation of the 10:1 batch-matching architecture, illustrated in **Figure 11**, the system exhibits a profound deviation from the global mean-field predictions. In the rationing-dominated regime (Scenario B), the ABM reveals that the "Never Submit" (NS) strategy unexpectedly overtakes the ecosystem. This occurs due to a phenomenon we term "Feasibility Collapse." Because the global publication slots (K_{store}) are strictly divided among the 100 journals, each local sub-market operates with an extremely narrow bottleneck. If even a small number of opportunistic (AS) authors flood a specific 10-person room, the local rationing probability (ρ) plummets below the economic break-even threshold required to justify the submission cost (c). Consequently, the "Defense by Congestion" becomes too punitive; even honest authors (OG) suffer negative expected payoffs and are forced to exit the market entirely, leaving NS as the only rational, risk-averse strategy.

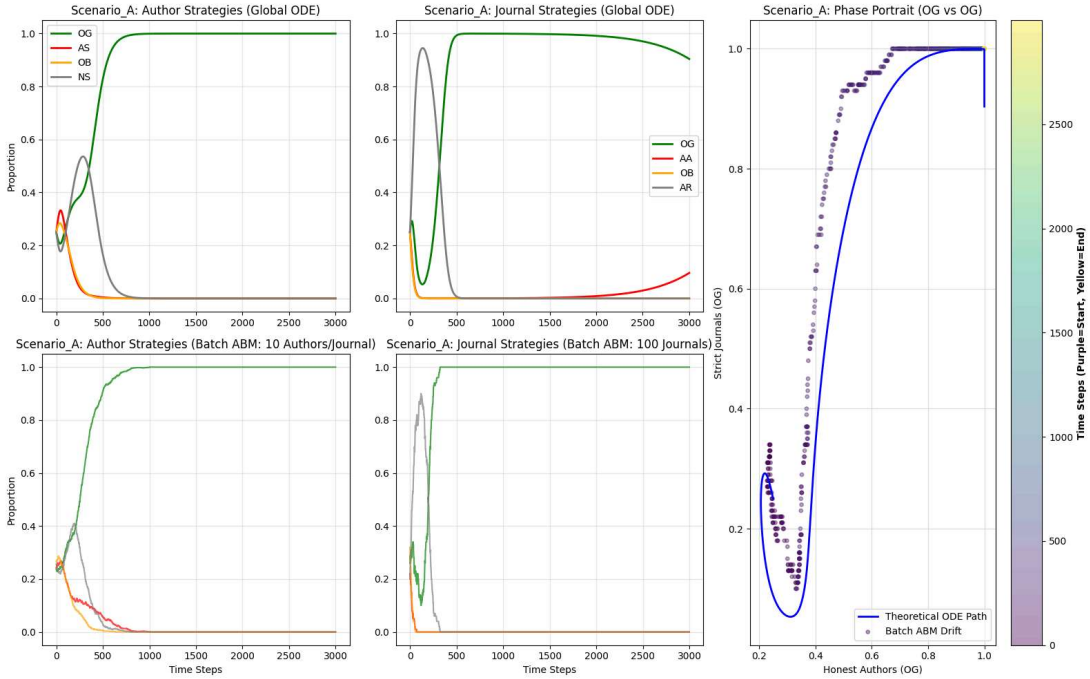


Figure 11: Co-evolutionary Dynamics under 10:1 Local Batch-Matching (Scenario B). The extreme fragmentation induces severe localized congestion variance, causing the "Never Submit" (NS) strategy to dominate due to Feasibility Collapse.

7 Conclusion

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